

Mohammad Marufur Rahman

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Research Interests

Machine Learning, Large Language Model, Computer Vision, AI in Healthcare, Internet of Things

Education

Wake Forest University, Winston-Salem, NC *August 2024 – May 2026*
MS in Computer Science
Khulna University of Engineering and Technology, Bangladesh *March 2016 – March 2020*
BS in Computer Science and Engineering

Experience

Graduate Research Assistant *Aug 2025 - May 2026*
Department of Computer Science, Wake Forest University, Winston-Salem, NC.
Advisor: Dr. Fan Yang
Graduate Student Mentor Training Course *Aug 2025 - Dec 2025*
Computing Research Association (CRA)
Graduate Teaching Assistant *Aug 2024 - May 2025*
Department of Computer Science, Wake Forest University, Winston-Salem, NC.
Lecturer *Dec 2020 – Aug 2024*
Department of Computer Science and Engineering
Ahsanullah University of Science and Technology, Dhaka, Bangladesh

Publications

- [Mohammad Marufur Rahman](#), Guanchu Wang, Kaixiong Zhou, Minghan Chen, Fan Yang, “Catastrophic Forgetting in Kolmogorov-Arnold Networks,” AAAI 2026 (Phase II, Under review).
- Hussam Al Daas, Grey Ballard, Md Taufique Hussain, Suraj Kumar, [Mohammad Marufur Rahman](#), Kathryn Rouse, “Nyström Low-Rank Approximation: Communication Lower Bounds and Efficient Algorithms.” IPDPS 2026 (Under review)
- [Mohammad Marufur Rahman](#), Md Milon Islam, Md MH Manik, Md Rabiul Islam, Mabrook S Al-Rakhami, “Machine learning approaches for tackling novel coronavirus (COVID-19) pandemic,” SN Computer Science.
- [Mohammad Marufur Rahman](#), S. Nooruddin, KM Azharul Hasan, NK Dey, “HOG+ CNN Net: Diagnosing COVID-19 and pneumonia by deep neural network from chest X-Ray images,” SN Computer Science.
- [Mohammad Marufur Rahman](#), Md MH Manik, Md Milon Islam, Saifuddin Mahmud, Jong-Hoon Kim, “An automated system to limit COVID-19 using facial mask detection in smart city network,” IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS).
- [Mohammad Marufur Rahman](#), Md Milon Islam, S. Ahmmed, SA Khan, “Obstacle and fall detection to guide the visually impaired people with real time monitoring,” SN Computer Science.
- [Mohammad Marufur Rahman](#), SA Khan, KM Azharul Hasan, “Word sense disambiguation by context detection,” EICT, 2019.

Projects

- Knowledge Editing** *Ongoing*
- Investigating a novel framework for parameter-efficient knowledge editing in large language models using factual data within a continual learning setting. *Advisor:* Dr. Fan Yang
- Parallel Nyström Approximation** [github](#) 
- Developed distributed and parallel algorithms for Nyström approximation of symmetric matrices; evaluated their performance on standard machine learning benchmark datasets. *Advisor:* Dr. Grey Ballard
- Human Activity Recognition (HAR)** [github](#) 
- Modified and evaluated sequence modeling architectures (Mamba, iTransformer, PatchTST) for activity classifi-

cation on the HAR70+ dataset of dual-sensor accelerometer recordings. *Tools Used:* Python, PyTorch.

Synthetic Data for Imbalanced Imaging Task

[github](#) 

- Investigated class imbalance issue in medical imaging. Implemented various augmentation techniques, including a diffusion model to generate synthetic image data. *Tools Used:* Python, PyTorch.

Parallel Matrix Multiplication

[github](#) 

- Implemented multiple matrix multiplication techniques in GPU and compared performance with MKL BLAS and cuBLAS. *Tools Used:* C++, CUDA.

Mutual Exclusion in Computer Network

[github](#) 

- Wrote a simple Shell program and implement Ricart and Agrawala's algorithm for mutual exclusion in computer networks. *Tools Used:* C

Technical Skills

Languages: C, C++, Java, Python, CUDA, MATLAB, LaTeX, HTML, CSS, JavaScript.

Frameworks: PyTorch, TensorFlow, Keras, Huggingface, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib.

Online Courses

- Machine Learning (Coursera).
- Mathematics for Machine Learning Specialization (Coursera).
- Deep Learning Specialization (Coursera).
- Machine Learning: Algorithms in the Real World Specialization (Coursera).